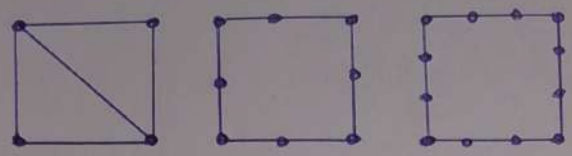
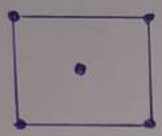
No.12 18-12-2024



узлы в не основной сетки



Эрмитовы КЭ

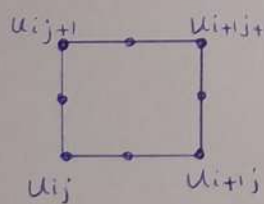
$$u,\ \underset{\underset{P_1}{\parallel}}{\frac{\partial u}{\partial x}},\ \underset{\underset{P_2}{\parallel}}{\frac{\partial u}{\partial y}}$$

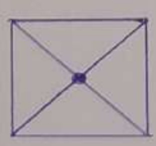
Эрмита

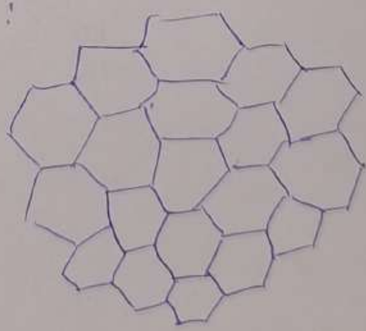
$$\left.\begin{array}{l} u_i,\ P_i \\ u_{i+1},\ P_{i+1} \end{array}\right\}\quad \begin{array}{l} P_3(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3 \\ P_3(x_i) = u_i,\quad P_3(x_{i+1}) = u_{i+1} \end{array}$$

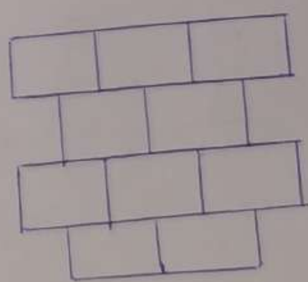
$$\left.\frac{dP_3}{dx}\right|_{x_i} = P_i \qquad \left.\frac{dP_3}{dx}\right|_{x_{i+1}} = P_{i+1}$$

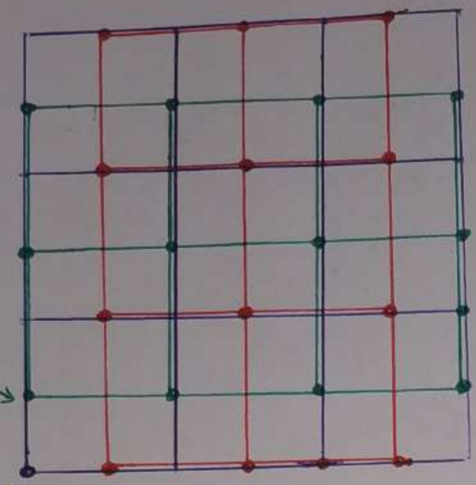
Сплайн дефекта 2!





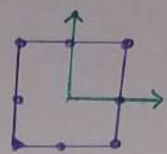






u_{ij}, v_{ij}, w_{ij}

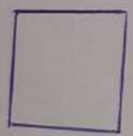
$$u = c_0 + c_1 x + c_2 y + c_3 x^2 + c_4 y^2 + c_5 xy + c_6 x y^2 + c_7 x^2 y$$

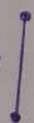


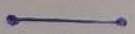
$$u_{ij} = u\left(-\frac{h_1}{2}, \frac{h_2}{2}\right) = c_0 - \frac{c_1 h_1}{2} - \frac{c_2 h_2}{2} + c_3 \frac{h_1^2}{4} + c_4 \frac{h_2^2}{4} + c_5 \frac{h_1 h_2}{2} - c_6 \frac{h_1 h_2^2}{8} - c_7 \frac{h_1^2 h_2}{8}$$

$$v_{ij} = u\left(0, -\frac{h_2}{2}\right) = c_0 - c_2 \frac{h_2}{2} + c_4 \frac{h_2^2}{4}$$

$$w_{ij} = u\left(-\frac{h_1}{2}, 0\right) = c_0 - c_1 \frac{h_1}{2} + c_3 \frac{h_1^2}{4}$$

u  $\rightarrow d_{0u}^+, d_{1u}^+, d_{2u}^+, d_{3u}^+$

v  $\rightarrow d_{0v}^+, d_{1v}^+$

 $\rightarrow d_{0w}^+, d_{1w}^+$

w

$\Rightarrow c_i = f(d_0^+ u, \dots, d_{1w}^+)$

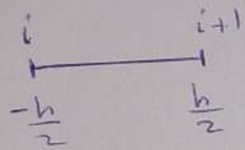
страница 32
ИММ_С
(2.53)

$$W = \int_a^b u^2 + \left(\frac{\partial u}{\partial x}\right)^2 dx \qquad u = c_0 + c_1 x + c_2 x^2 + c_3 x^3$$

$$P = \frac{du}{dx} = c_1 + 2c_2 x + 3c_3 x^2$$

$$d_0^+ u = \frac{1}{2}(u_i + u_{i+1}) = c_0 + c_2 \frac{h^2}{4}$$

$$d_1^+ u = \frac{1}{h}(u_{i+1} - u_i) = c_1 + c_3 \frac{h^2}{4}$$



$$d_0^+ P = \frac{1}{2}(P_i + P_{i+1}) = c_1 + \frac{3}{4} c_3 h^2$$

$$d_1^+ P = \frac{1}{h}(P_{i+1} - P_i) = 2c_2$$